

2 Generating functions

Given a sequence $\{F_n\}_{n=0}^\infty$, which may arise either combinatorially, recursively, or in terms of some number-theoretic law, an important tool in its study is the passage to its **generating function**, defined by

$$F(x) = \sum_{n=0}^{\infty} F_n x^n.$$

Often times, the defining properties of the sequence $\{F_n\}$ imply interesting algebraic or analytic properties of the function $F(x)$, and exploiting these can eventually lead us back to new insights about the sequence $\{F_n\}$. A very simple-minded example is given by the Fibonacci sequence. (See Exercise 2). Here we want to study less elementary examples of this idea, related to the Θ function.

We shall first discuss very briefly the theory of partitions.

The **partition function** is defined as follows: if n is a positive integer, we let $p(n)$ denote the numbers of ways n can be written as a sum of positive integers. For instance, $p(1) = 1$, and $p(2) = 2$ since $2 = 2 + 0 = 1 + 1$. Also, $p(3) = 3$ since $3 = 3 + 0 = 2 + 1 = 1 + 1 + 1$. We set $p(0) = 1$ and collect some further values of $p(n)$ in the following table.

n	0	1	2	3	4	5	6	7	8	$\dots$	12
$p(n)$	1	1	2	3	5	7	11	15	22	$\dots$	77

The first theorem is Euler's identity for the generating function of the partition sequence $\{p(n)\}$, which is reminiscent of the product formula for the zeta function.

Theorem 2.1 *If $|x| < 1$, then $\sum_{n=0}^{\infty} p(n)x^n = \prod_{k=1}^{\infty} \frac{1}{1-x^k}$.*

Formally, we can write each fraction as

$$\frac{1}{1-x^k} = \sum_{m=0}^{\infty} x^{km},$$

and multiply these out together to obtain $p(n)$ as the coefficient of x^n . Indeed, when we group together equal integers in a partition of n , this partition can be written as

$$n = m_1 k_1 + \dots + m_r k_r,$$

where $k_1, \dots, k_r$ are distinct positive integers. This partition corresponds to the term

$$(x^{k_1})^{m_1} \dots (x^{k_r})^{m_r}$$

that arises in the product.

The justification of this formal argument proceeds as in the proof of the product formula for the zeta function (Section 1, Chapter 7); this is based on the convergence of the product $\prod 1/(1 - x^k)$. This convergence in turn follows from the fact that for each fixed $|x| < 1$ one has

$$\frac{1}{1 - x^k} = 1 + O(x^k).$$

A similar argument shows that the product $\prod 1/(1 - x^{2n-1})$ is equal to the generating function for $p_o(n)$, the number of partitions of n into odd parts. Also, $\prod (1 + x^n)$ is the generating function for $p_u(n)$, the number of partitions of n into unequal parts. Remarkably, $p_o(n) = p_u(n)$ for all n , and this translates into the identity

$$\prod_{n=1}^{\infty} \left(\frac{1}{1 - x^{2n-1}} \right) = \prod_{n=1}^{\infty} (1 + x^n).$$

To prove this note that $(1 + x^n)(1 - x^n) = 1 - x^{2n}$, and therefore

$$\prod_{n=1}^{\infty} (1 + x^n) \prod_{n=1}^{\infty} (1 - x^n) = \prod_{n=1}^{\infty} (1 - x^{2n}).$$

Moreover, taking into account the parity of integers, we have

$$\prod_{n=1}^{\infty} (1 - x^{2n}) \prod_{n=1}^{\infty} (1 - x^{2n-1}) = \prod_{n=1}^{\infty} (1 - x^n),$$

which combined with the above proves the desired identity.

The proposition that follows is deeper, and in fact involves the Θ function directly. Let $p_{e,u}(n)$ denote the number of partitions of n into an even number of unequal parts, and $p_{o,u}(n)$ the number of partitions of n into an odd number of unequal parts. Then, Euler proved that, unless n is a pentagonal number, one has $p_{e,u}(n) = p_{o,u}(n)$. By definition, **pentagonal numbers**² are integers n of the form $k(3k + 1)/2$, with $k \in \mathbb{Z}$. For

²The traditional definition is as follows. Integers of the form $n = k(k - 1)/2$, $k \in \mathbb{Z}$, are “triangular numbers”; those of the form $n = k^2$ are “squares”; and those of the form $k(3k + 1)/2$ are “pentagonal numbers.” In general, numbers of the form $(k/2)((\ell - 2)k + \ell - 4)$ are associated with an ℓ -sided polygon.

example, the first few pentagonal numbers are 1, 2, 5, 7, 12, 15, 22, 26, ... In fact, if n is pentagonal, then

$$p_{e,u}(n) - p_{o,u}(n) = (-1)^k, \quad \text{if } n = k(3k+1)/2.$$

To prove this result, we first observe that

$$\prod_{n=1}^{\infty} (1 - x^n) = \sum_{n=1}^{\infty} [p_{e,u}(n) - p_{o,u}(n)] x^n.$$

This follows since multiplying the terms in the product, we obtain terms of the form $(-1)^r x^{n_1 + \dots + n_r}$ where the integers $n_1, \dots, n_r$ are distinct. Hence in the coefficient of x^n , each partition $n_1 + \dots + n_r$ of n into an even number of unequal parts contributes for $+1$ (r is even), and each partition into an odd number of unequal parts contributes -1 (r is odd). This gives precisely the coefficient $p_{e,u}(n) - p_{o,u}(n)$.

With the above identity, we see that Euler's theorem is a consequence of the following proposition.

Proposition 2.2
$$\prod_{n=1}^{\infty} (1 - x^n) = \sum_{k=-\infty}^{\infty} (-1)^k x^{\frac{k(3k+1)}{2}}.$$

Proof. If we set $x = e^{2\pi i u}$, then we can write

$$\prod_{n=1}^{\infty} (1 - x^n) = \prod_{n=1}^{\infty} (1 - e^{2\pi i n u})$$

in terms of the triple product

$$\prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1}e^{2\pi i z})(1 + q^{2n-1}e^{-2\pi i z})$$

by letting $q = e^{3\pi i u}$ and $z = 1/2 + u/2$. This is because

$$\prod_{n=1}^{\infty} (1 - e^{2\pi i 3nu})(1 - e^{2\pi i (3n-1)u})(1 - e^{2\pi i (3n-2)u}) = \prod_{n=1}^{\infty} (1 - e^{2\pi i nu}).$$

By Theorem 1.3 the product equals

$$\sum_{n=-\infty}^{\infty} e^{3\pi i n^2 u} (-1)^n e^{2\pi i n u/2} = \sum_{n=-\infty}^{\infty} (-1)^n e^{\pi i n(3n+1)u}$$

$$= \sum_{n=-\infty}^{\infty} (-1)^n x^{n(3n+1)/2},$$

which was to be proved.

We make a final comment about the partition function $p(n)$. The nature of its growth as $n \rightarrow \infty$ can be analyzed in terms of the behavior of $1/\prod_{n=1}^{\infty} (1-x)^n$ as $|x| \rightarrow 1$. In fact, by elementary considerations, we can get a rough order of growth of $p(n)$ from the growth of the generating function as $x \rightarrow 1$; see Exercises 5 and 6. A more refined analysis requires the transformation properties of the generating function which goes back to the corresponding Proposition 1.9 about η . This leads to a very good asymptotic formula for $p(n)$. It may be found in Appendix A.

3 The theorems about sums of squares

The ancient Greeks were fascinated by triples of integers (a, b, c) that occurred as sides of right triangles. These are the “Pythagorean triples,” which satisfy $a^2 + b^2 = c^2$. According to Diophantus of Alexandria (ca. 250 AD), if c is an integer of the above kind, and a and b have no common factors (a case to which one may easily reduce), then c is the sum of two squares, that is, $c = m^2 + n^2$ with $m, n \in \mathbb{Z}$; and conversely, any such c arises as the hypotenuse of a triangle whose sides are given by a Pythagorean triple (a, b, c) . (See Exercise 8.) Therefore, it is natural to ask the following question: which integers can be written as the sum of two squares? It is easy to see that no number of the form $4k + 3$ can be so written, but to determine which integers can be expressed in this way is not obvious.

Let us pose the question in a more quantitative form. We define $r_2(n)$ to be the number of ways n can be written as the sum of two squares, counting obvious repetitions; that is, $r_2(n)$ is the number of pairs (x, y) , $x, y \in \mathbb{Z}$, so that

$$n = x^2 + y^2.$$

For example, $r_2(3) = 0$, but $r_2(5) = 8$ because $5 = (\pm 2)^2 + (\pm 1)^2$, and also $5 = (\pm 1)^2 + (\pm 2)^2$. Hence, our first problem can be posed as follows:

Sum of two squares: Which integers can be written as a sum of two squares? More precisely, can one determine an expression for $r_2(n)$?

Next, since not every positive integer can be expressed as the sum of two squares, we may ask if three squares, or possibly four squares suffice.